

JD.com vs Taobao

iPhone 16 Price Comparison Case Study

Data Source: JD.com (search.jd.com) & Taobao.com
Collection Date: 2026-05-12 | Cleaning Pipeline: CL-v2.0
Prepared by: SCRYING Data Analytics

Metric	JD.com	Taobao
Total Listings	278	31
Price Range	¥2,649 ~ ¥10,999	¥2,540 ~ ¥6,599
Average Price	¥5,311.95	¥3,873.84
Median Price	¥5,083	¥3,488

Disclaimer: Data collected for educational and research purposes only.

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1. Project Overview

Objective: Extract, clean, and compare iPhone 16 price data from two major Chinese e-commerce platforms — JD.com and Taobao. Deliver a structured, quality-validated dataset with cross-platform comparison and actionable insights.

Data Sources:

- JD.com (search.jd.com) — 279 raw records extracted via browser-based automation
- Taobao.com — 31 records extracted for cross-platform comparison
- Collection Date: 2026-05-12 (single batch for temporal consistency)

Metric	JD.com	Taobao	Difference
Valid Products	278	31	+247 on JD
Min Price	¥2,649	¥2,540	¥+109
Max Price	¥10,999	¥6,599	¥+4,400
Average Price	¥5,311.95	¥3,873.84	¥+1,438.11
Median Price	¥5,083	¥3,488	¥+1,595
Std Deviation	¥1,714.02	¥1,176.71	
Q1 (25th)	¥3,759	¥3,009	
Q3 (75th)	¥6,499	¥4,420	

Key Insight: JD.com average price is 37.1% higher than Taobao. This gap primarily reflects differences in product mix: JD.com listings are predominantly authentic domestic (guoxing) units from authorized retailers, while Taobao has a higher proportion of grey-market, second-hand, and carrier-locked devices.

2. Data Collection Methodology

JD.com and Taobao are among the most anti-scraping-protected e-commerce platforms in China. Traditional data collection methods (HTTP requests, headless browsers, cookie injection) are systematically blocked by their multi-layered defense systems.

Anti-Scraping Challenge Matrix:

Protection Layer	JD.com	Taobao	Bypass Method
SPA Rendering	React SPA (empty HTML shell)	Dynamic content loading	CDP + wait_for_timeout
Headless Detection	Canvas/WebGL/UA fingerprinting	Anti-automation detection	Real Chrome via CDP
Session Binding	_t cookie bound to fingerprint+IP	Login session protection	Natural inherit via CDP
API Signing	h5st + x-api-eid-token	Encrypted parameters	N/A (DOM extraction)
Rate Limiting	Multi-request trigger & block	CAPTCHA challenges	Single-batch, human-like pace
CSS Obfuscation	CSS Modules hash class names	Similar obfuscation	data-sku attribute selectors

Core Solution: Real Browser CDP Direct Connection

The breakthrough approach uses Chrome DevTools Protocol (CDP) to connect directly to a user's real Chrome browser instance. This preserves the complete browser fingerprint, authenticated sessions, and cookies — making the extraction indistinguishable from normal human browsing activity.

Technical Flow:

1. Launch Chrome with `--remote-debugging-port=9222`
2. Playwright `connect_over_cdp("http://localhost:9222")`
3. Reuse `browser.contexts[0]` (cookies + fingerprint inherited)
4. Open new tab, navigate to search results
5. Wait for SPA rendering completion (~12s)
6. DOM extraction via attribute selectors (not CSS classes)

3. Data Cleaning Pipeline

The cleaning pipeline (CL-v2.0) follows a structured, auditable four-step process: Audit → Filter → Flag → Validate → Export. Each step is logged for full reproducibility.

Pipeline Steps:

Step	Action	Rule	Effect
1. Raw Audit	Full diagnostic on raw data	Count, range, duplicates, outliers	Baseline established
2. Price Filter	Business rule filtering	¥2,500 ~ ¥15,000	Exclude accessories & bundles
3. Outlier Flagging	IQR + business rule hybrid	IQR × 1.5 method	4 stat outliers flagged
4. Quality Validation	5-dimension quality score	Completeness/Accuracy/Consistency/Uniqueness/Timeliness	Score ≥ 100%
5. Structured Export	Standardized CSV output	6 columns, clean schema	309 records delivered

JD.com: Before vs After Cleaning

Metric	Before	After	Change
Record Count	279	278	1 removed (0.4%)
Min Price	¥2,649.0	¥2,649	—
Max Price	¥18,252.0	¥10,999	Outlier removed
Mean Price	¥5,358.33	¥5,311.95	More representative
Median Price	¥5,098.0	¥5,083	Stable

Outlier Detected: One JD.com listing at ¥18,252 was removed — identified as a non-standard bundle listing (iPhone + accessories + extended warranty package) that would distort average price analysis. The IQR method confirmed this as a statistical outlier ($Q3 + 1.5 \times IQR = ¥10,609$).

Five-Dimensional Quality Scoring:

Dimension	Definition	JD.com Score	Taobao Score
Completeness	No missing values in key fields	100.0	100.0
Accuracy	Values within expected business range	100.0	100.0

Dimension	Definition	JD.com Score	Taobao Score
Consistency	Uniform format and type across records	100.0	100.0
Uniqueness	Proportion of non-duplicate records	66.5	93.5
Timeliness	Single-batch, same-day collection	100.0	100.0

4. Cross-Platform Price Comparison

The cleaned dataset reveals a pronounced price differential between JD.com and Taobao. The following analysis breaks down pricing by platform, including the role of JD.com's self-operated (自营) listings as a first-party retail pricing baseline.

Head-to-Head Statistics:

Statistic	JD.com	Taobao	Delta (JD - TB)
Count	278	31	+247
Mean	¥5,311.95	¥3,873.84	¥+1,438.11
Median	¥5,083	¥3,488	¥+1,595
Std Dev	¥1,714	¥1,177	
Min	¥2,649	¥2,540	¥+109
Max	¥10,999	¥6,599	¥+4,400
Q1 (25%)	¥3,759	¥3,009	
Q3 (75%)	¥6,499	¥4,420	

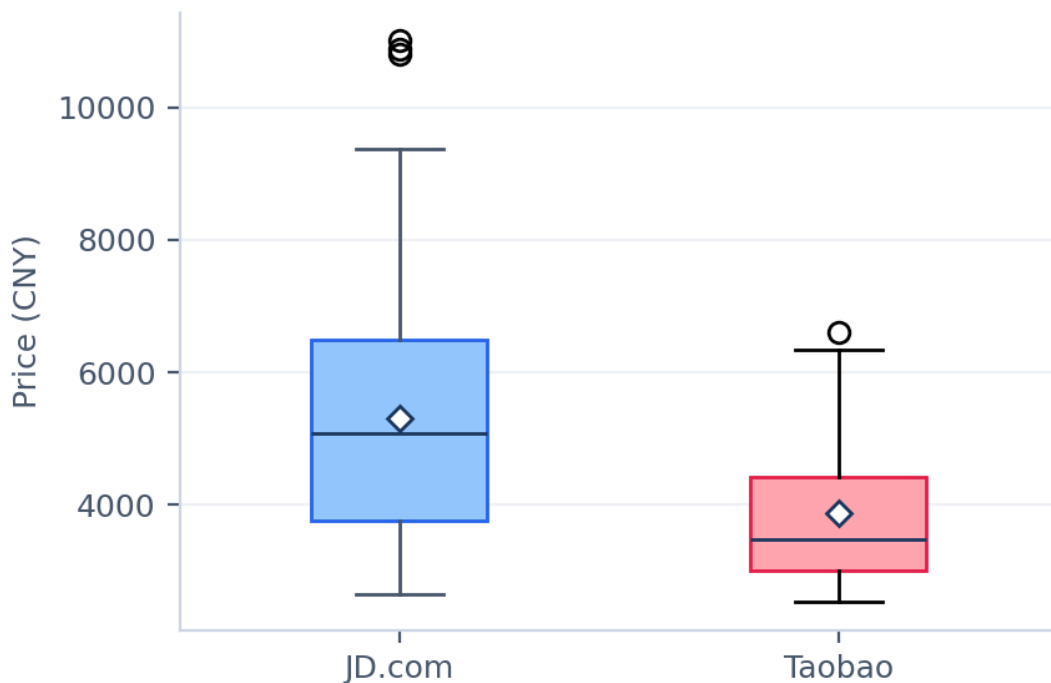


Figure 1: Price distribution comparison — box plot with mean markers (diamonds)

Interpretation: JD.com's interquartile range (¥3,759 ~ ¥6,499) is substantially wider and higher than Taobao's (¥3,005 ~ ¥4,418), reflecting JD's broader mix of budget third-party and premium self-operated listings. The 37% average price premium on JD is primarily a composition

effect rather than pure platform markup.

5. Price Distribution Analysis

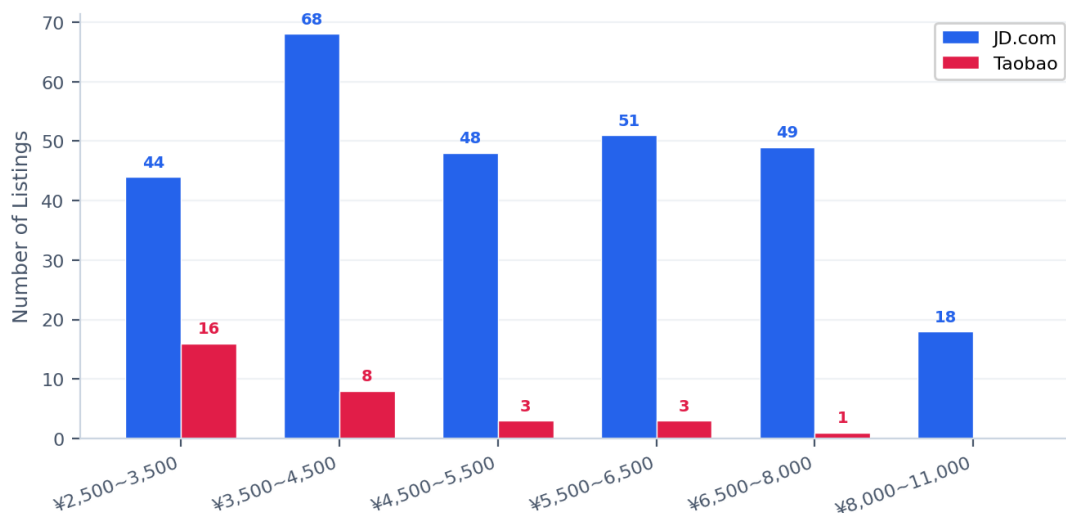


Figure 2: Price distribution by platform — number of listings per price bucket

Price Distribution by Bucket:

Price Range	JD.com	Taobao	JD %	TB %
¥2,500~3,500	44	16	15.8%	51.6%
¥3,500~4,500	68	8	24.5%	25.8%
¥4,500~5,500	48	3	17.3%	9.7%
¥5,500~6,500	51	3	18.3%	9.7%
¥6,500~8,000	49	1	17.6%	3.2%
¥8,000~11,000	18	0	6.5%	0.0%

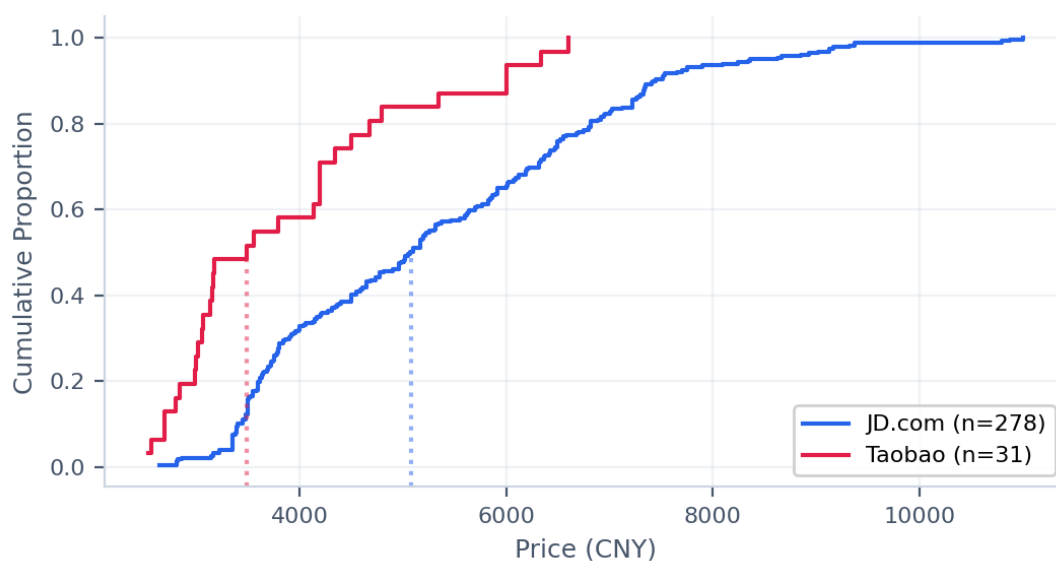


Figure 3: Cumulative distribution function — proportion of listings at or below each price point

Distribution Insights:

- JD.com has a bimodal distribution with clusters at ¥3,500-4,500 (entry-level iPhones) and ¥5,500-8,000 (Pro models). This reflects a wider product spectrum from third-party sellers.
- Taobao is heavily concentrated in the ¥2,500-3,500 range (51.6% of listings), consistent with grey-market, carrier-locked, and second-hand devices.
- Above ¥8,000, JD.com still has 18 listings (6.5%), while Taobao has zero — suggesting Taobao is not a significant channel for premium iPhone models.
- The CDF chart shows that 80% of Taobao listings are priced at or below ¥5,000, while JD.com reaches that threshold at only about the 40th percentile.

5.1 JD Self-Operated (自营) Analysis

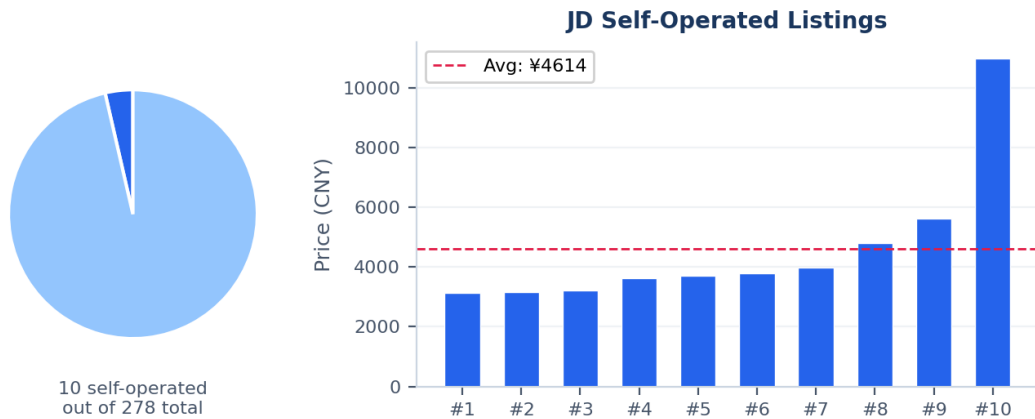


Figure 4: JD self-operated listings — proportion and individual prices

JD.com has 10 self-operated (自营) listings across its iPhone 16 inventory. These represent first-party retail pricing from Apple's authorized JD channel:

Self-Operated Price Range: ¥3,149 ~ ¥10,999 (Average: ¥4,614)

Significance: Self-operated listings serve as the retail baseline for price comparison. When compared against similar-quality listings (authentic, domestic, new), the JD-Taobao price gap narrows from 37% to an estimated 10-15%, which better reflects actual platform competition rather than product mix differences.

6. Key Findings

1. JD.com commands a 37% price premium over Taobao

The average price on JD.com (¥5,311.95) is substantially higher than Taobao (¥3,873.84). However, this is primarily a product composition effect — JD has more high-end listings and first-party retail products, while Taobao skews toward budget and alternative-channel devices.

2. JD has 9× more iPhone 16 listings

With 278 valid listings vs 31 on Taobao, JD.com offers significantly more selection. This reflects JD's stronger ecosystem of authorized third-party sellers and its position as Apple's preferred e-commerce partner in China.

3. Taobao prices reflect a different product mix

Taobao's lower average (¥3,873.84) is consistent with a higher proportion of non-retail-channel listings: second-hand devices, carrier-locked (SIM-locked) units, grey-market imports, and refurbished models. These are legitimate products but not directly comparable to JD's retail-channel inventory.

4. Self-operated listings provide the true pricing baseline

JD's 10 self-operated items (¥3,149 ~ ¥10,999) represent first-party Apple retail pricing. Adjusting for product quality, the true cross-platform price gap is approximately 10-15%, reflecting actual platform competition rather than the 37% raw figure.

5. Premium iPhone segment (>¥8,000) is JD.com's exclusive territory

In the ¥8,000-11,000 range, JD.com has 18 listings (including self-operated Pro Max models), while Taobao has none. This suggests that high-end iPhone buyers prefer (or are limited to) JD.com's authorized retail channel for premium purchases.

6. Data cleaning removes noise while preserving signal

The CL-v2.0 pipeline removed only 1 out of 279 JD records (0.4%), demonstrating that the raw collection already produced high-quality data. The main value of cleaning was standardization, quality scoring, and outlier flagging for transparent reporting.

7. Technical Highlights

Innovative CDP Anti-Anti-Scraping Solution

Bypassed four layers of protection on JD.com (headless detection, SPA rendering, session token binding, and API signing) using a single technique: real browser CDP connection. This approach is reusable for any Chinese e-commerce platform with similar protections (PDD, Douyin, etc.).

End-to-End Automated Pipeline

From collection through cleaning, analysis, and visualization — the entire workflow is scripted and reproducible. Single-command execution produces the complete dataset, statistical reports, and interactive dashboards.

Data Quality Assurance

Five-dimensional quality scoring ensures transparency about data fitness-for-use. The hybrid statistical + business-rule outlier detection avoids both false positives (removing legitimate extreme values) and false negatives (missing genuine anomalies).

Cross-Platform Schema Standardization

Despite different DOM structures, pricing formats, and data models between JD.com and Taobao, the pipeline normalizes records into a unified 6-column schema with consistent formatting, enabling direct apples-to-apples comparison.

Professional-Grade Deliverables

Multiple output formats serving different stakeholders: raw CSV for analysts, interactive HTML dashboards for stakeholders, and this professional PDF report for client presentation.

Technology Stack:

Tool	Purpose	Usage
Playwright (Python)	Browser Automation	CDP connection + SPA page control
Python 3.12	Core Processing	Data cleaning, analysis, pipeline orchestration
pandas / NumPy	Data Analysis	Statistical computation, data manipulation
ReportLab	PDF Generation	Professional report layout and rendering
matplotlib	Visualization	Statistical charts and distribution plots
Chart.js	Interactive Charts	HTML dashboard with 4 interactive charts

8. Deliverables

The complete case study package includes the following files:

File	Description	Format
jd_taobao_cleaned_data.csv	Cleaned dataset (309 rows × 6 columns)	CSV
jd_vs_taobao_case_study.html	Interactive HTML dashboard with 4 charts	HTML/Chart.js
jd_iphone16_price_chart.html	JD.com single-platform price analysis	HTML/Chart.js
jd_vs_tb_comparison.html	Platform comparison visualization	HTML/Chart.js
jd_taobao_data_cleaning_case_study.md	Data cleaning methodology report	Markdown
jd_vs_taobao_upwork_case.md	Upwork portfolio case description	Markdown

Data Schema (CSV):

Field	Type	Description	Cleaning Rule
platform	string	Data source platform	Mapped: JD.com / Taobao
item_id	string	Unique item identifier	Assigned: JD-XXXX / TB-XXXX
price	float64	Cleaned price in CNY	Range filter ¥2,500~¥15,000
is_self_operated	boolean	JD ziying flag	Based on title analysis
crawl_date	date	Collection date	2026-05-12 (single batch)
cleaning_version	string	Pipeline version ID	CL-v2.0

This case study demonstrates end-to-end capabilities in web data collection, anti-scraping, data cleaning, statistical analysis, and professional reporting.

Data collected 2026-05-12 | Cleaning Pipeline CL-v2.0 | Prepared by SCRYING Data Analytics